

CE 310 — Week 2 Assignment: Tails, Thresholds, and Design Choices

Due: Wednesday of Week 3 by 9:00 AM — submit to D2L

Overview

This assignment extends the Week 2 lab into new territory. You will compute percentile thresholds that were not in the lab, compare design implications across different exceedance levels, and write a professional interpretation of why distribution tails — not averages — drive engineering design.

How to Submit

- ☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Major declaration	Enter CE or ArcE in cell C6 — this determines which Problem 2 track is graded
Column layout	Enter your formula in Column C, any starred (★) intermediate value in Column D, and the final answer in Column E — do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. Screenshots are not accepted
Written responses	Type them in the yellow cells lower down the same sheet — do not submit a separate PDF
File name	CE310_W2_<NetID>.xlsx e.g. CE310_W2_abc123.xlsx

Requirement	Detail
Submission	Upload the .xlsx file to D2L Dropbox before the deadline

Problem 1 — Cold-Weather Thresholds and Heating Design (30 pts)

The Week 2 lab focused on the hot end of Tucson's temperature distribution (95th percentile cooling). This problem addresses the cold end — relevant for heating system design and freeze protection.

1a. How many hours in 2023 did Tucson's temperature fall below each threshold? Complete the table.

Threshold (°F)	Hours Below	% of Year
40		
32 (freezing)		

Use COUNTIF for each threshold. Show your formula in your submission.

Your lab did not compute any below-threshold counts — the lab focused on the hot tail (>95°F, >100°F). These are new computations.

1b. The 5th percentile of temperature is the heating design analog to the ASHRAE 1% cooling design temperature from Week 1. It represents the temperature below which only the coldest 5% of hours fall.

Calculate the 5th percentile of temperature using PERCENTILE.

A small commercial building in Tucson has a heat loss rate of $Q = U \times A \times \Delta T$, where: - $U = 0.4 \text{ BTU}/(\text{hr} \cdot \text{ft}^2 \cdot ^\circ\text{F})$ (overall thermal conductance) - $A = 5,000 \text{ ft}^2$ (building envelope area) - $\Delta T = \text{indoor setpoint } (70^\circ\text{F}) \text{ minus outdoor design temperature}$

Calculate the heating load (BTU/hr) for: - Approach A (wrong): Size the heating system for the annual average temperature (72°F). - Approach B (correct): Size for the 5th percentile temperature.

Note for Approach A: Tucson's annual average temperature (72°F) is higher than the indoor heating setpoint (70°F). This means $\Delta T = 70 - 72 = -2^\circ\text{F}$, and the formula gives a negative heating load — the math implies the building is losing heat to the outdoors (i.e., cooling is needed, not heating). A negative result is not a calculation error. It is the point: average-based design says no heating system is needed at all, which is demonstrably wrong on winter mornings. Report the negative value, then contrast it with Approach B.

Find the load difference. In 2–3 sentences, explain what happens to building occupants on the coldest mornings of the year if the installed system can only handle Approach A’s capacity.

Enter your response in the P1b_written cell of your Answer Sheet.

1c. Your lab computed the temperature IQR ($B3 \approx 28^\circ\text{F}$). Calculate the lower outlier fence for temperature:

Lower fence = $Q1 - 1.5 \times \text{IQR}$

How many hours in 2023 fell below this lower fence? Use COUNTIF. In 1–2 sentences, interpret whether these hours qualify as statistical outliers in Tucson’s climate context, and whether an engineer should still design for them.

Enter your response in the P1c_written cell of your Answer Sheet.

Problem 2 — Engineering Interpretation (18 pts) — answer ONLY your track

Declare CE or ArcE in cell C6 of your Answer Sheet. That declaration determines which track below is graded — complete only its rows and leave the other track’s rows blank.

Track A — Civil Engineering: Wind Design Percentile Comparison

Your lab computed the 95th percentile wind speed ($B10 \approx 17.2$ mph) and the wind pressure it produces ($B11 \approx 0.985$ psf), using the ASCE 7 velocity pressure relationship:

$$q = 0.00256 \cdot K \cdot V^2$$

with $K = 1.3$ (a combined exposure and gust factor) and V in mph. This track moves to a more demanding design level.

2A-a. Compute all three of the following:

Row	Quantity
2A-a-i	99th percentile wind speed, using =PERCENTILE(C2:C8761,0.99)
2A-a-ii	Wind pressure at that 99th percentile speed, using the formula above
2A-b	The ratio of the 99th percentile pressure to your lab’s 95th percentile pressure (B11)

Note: Wind pressure grows with the square of speed. Before you compute the ratio in 2A-b, predict it from the speed ratio alone — then check whether your computed pressure ratio matches that prediction. The relationship between the two is the point of this part.

2A-c. By definition the 99th percentile is exceeded about 1% of the time and the 95th about 5% — roughly 87 versus 435 of the 8,700 hours with valid wind readings. In 2–3 sentences, explain what your 2A-b ratio means for a designer choosing between the two levels: how much additional structural capacity does moving from the 95th to the 99th percentile demand, and is that proportional to the difference in exceedance hours?

2A-d. Reference ASCE 7-22's use of design wind speeds tied to risk categories. In 2–3 sentences, explain why a hospital and a storage shed are not designed to the same exceedance probability, using your two pressure values as the concrete illustration.

Track A students: enter both written responses in the P2_CE_written cell of your Answer Sheet.

Track B — Architectural Engineering: Seasonal Humidity Analysis

Your lab computed the annual humidity 90th percentile (B12) and August average humidity (B13 $\approx$ 39.95%). This track adds a winter-vs-summer seasonal breakdown and a summer design value.

2B-a. Compute all three of the following:

Row	Quantity
2B-a-i	Winter average relative humidity — December (12), January (1), February (2)
2B-a-ii	Summer average relative humidity — June (6), July (7), August (8)
2B-a-iii	Summer 90th percentile relative humidity — the design value for 2B-c

For the two seasonal averages, use three separate AVERAGEIF calls, one per month, then average the three results:

$$=(\text{AVERAGEIF}(E2:E8761,12,D2:D8761) + \text{AVERAGEIF}(E2:E8761,1,D2:D8761) + \text{AVERAGEIF}(E2:E8761,2,D2:D8761)) / 3$$

Note: Averaging the three monthly means is not quite the same as pooling all winter hours, because the months differ in length. Either method is accepted; the two answers differ by less than 0.1 percentage points.

For 2B-a-iii, restrict the percentile to summer hours only. The most direct route is a helper column that copies the humidity value when the month is 6, 7 or 8 and is blank otherwise, then =PERCENTILE() over that column.

2B-b. Your results will show that winter average humidity is higher than summer average humidity in Tucson — the opposite of what most students expect. In 2–3 sentences, explain this counterintuitive result. Reference the North American Monsoon and the role of Pacific moisture in Tucson winters.

2B-c. Despite winter having the higher average, ASHRAE engineers size cooling-season dehumidification for summer conditions. Compare your 2B-a-iii summer 90th percentile against your 2B-a-i winter average, and in 2–3 sentences explain why the summer design percentile — not the higher winter mean — governs equipment selection. Reference ASHRAE 62.1 and the concept of design conditions.

Track B students: enter both written responses in the P2_ArcE_written cell of your Answer Sheet.

Problem 3 — Reflection (12 pts)

Answer in one paragraph (100–150 words):

A colleague argues: “Choosing the 95th vs. 99th percentile as the design threshold is a minor technical detail that engineers debate too much — the difference is only a few percent of hours, so the practical impact on design must be small.”

Using at least one specific number from your analysis (a pressure ratio, an exceedance count, or a computed threshold), write a professional response that either agrees, disagrees, or qualifies this argument. Address whether the “few percent of hours” framing is the right way to evaluate the impact of the design choice.

Enter your response in the P3_written cell of your Answer Sheet.

Submission Requirements

- Type all written responses in the designated yellow cells of your Assignment Answer Sheet. Submit CE310_W2_<NetID>.xlsx to D2L.
 - Type every formula as text in Col C (Approach / Formula), exactly as you entered it in Excel. Screenshots are not accepted.
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Grading Summary

Problem	Points
Problem 1a — 2 cold thresholds with COUNTIF formulas (4.5 pts each)	9
Problem 1b — Heating load comparison + written explanation	12
Problem 1c — Lower outlier fence + interpretation	9
Problem 2 — Engineering Interpretation, by track (Track A wind percentiles / Track B seasonal humidity)	18
Problem 3 — Reflection paragraph	12
Total	60 pts

This assignment is worth 60 raw points and will be scaled to the Post-Lab Memo category in the course gradebook.